Urja Jain

STUDENT, RESEARCHER, FOUNDER

I am a motivated freshman at UC Berkeley passionate about solving global challenges at the intersection of medicine, technology, and society. Through nanomedicine research at Albert Einstein College of Medicine and cardiovascular shadowing, I developed a strong foundation in healthcare innovation. Leading non-profits and community initiatives has equipped me with the ability to navigate diverse perspectives and implement solutions that address inequity and sustainability. I am especially interested in consulting as a way to translate biotechnology and healthcare advancements into strategies that improve access to care, strengthen health systems, and promote sustainable growth worldwide.

LEADERSHIP SKILLS

Critical Problem-Solving
Collaboration
Organization

Bridge Builder
Innovation
Creativity

RESEARCH SKILLS

ELISA Assay
Cell Culture
Nanoparticle Chemistry

Spectrophotometry
ImageJ Analysis
Tissue Engineering

EXTRACURRICULAR ACTIVITIES

DECA (Distributive Education Clubs of America) **September 2021- Present**
Medalist, President Emeritus, and NY Regional Representative

- Created several projects combining AI, medicine, and bioengineering at DECA conferences; presented an award-winning project on inventing a clinical biomarker for pain
- Planned and led weekly board meetings and oversaw competitive events; created a chapter mentorship program; tripled membership under my leadership
- Led DECA at the regional level and organized conferences in my role as the Region 3 Board Representative

Pre-Medical Club **June 2022 - Present**
Founder and President

- Facilitated weekly bioethical debates, inspired high school students to create passion projects relating to medical interests, led fundraising initiatives, and organized medical field trips
- Competed in the Medical Disparities event for HOSA

Table Tennis Club **September 2021- Present**
Founder and President

- Created a student-led organization aimed at reducing the mental health impact of school work on teenagers at Ardsley High School through playful competition, music, and fundraisers

FIRST Robotics **June 2021- August 2022**
Head of Outreach

- Responsible for team outreach in the community and communication; creation of social media and competition materials; organized fundraising events raising hundreds of dollars; executed Python classes for elementary school children; program was recognized with the Motivate Award

COMMUNITY SERVICE

Westchester Durga Temple **March 2024 - Present**
Volunteer and Project Leader

- Organized the collection of Indian clothes from Indian-American families in the community
- Delivered clothes to underprivileged communities and schools in India
- Coordinated the delivery of menstrual hygiene products in rural communities

Bake Back America **January 2022 - Present**
Curriculum Leader, Writer, and Founder

- Created a weekly medical program for elementary school students in a homeless shelter, teaching them about careers and projects in science and medicine; responsible for curriculum development and creation of resources; authored children's booklets that explain scientific projects integrated with bioethics, disparities, mental health, and women's health, and distributed nationally
- Mentored multiple new volunteers, teaching them valuable communication and leadership skills
- Organized, and led initiatives at Columbia University working with inter-city high school students; responsible for fundraising
- Collaboratively created curriculum and led student-based financial literacy initiatives

RESEARCH EXPERIENCE

Albert Einstein College of Medicine **March 2023 - Present**
Student Researcher

- Project I:
  - Researched and developed a novel therapeutic for the accessible and sustainable treatment of diabetic wounds combining capsaicin and metformin nanoparticles using green chemistry.
  - Co-authored a paper in the European Heart Journal on a novel cardiovascular therapy drug, aprocitentan, and its role in alleviating mitochondrial stress
  - Collaborated and problem-solved, as a high school student, with scientists from other departments
  - Planned and executed experiments in formulating chitosan nanoparticles; characterizing them for morphology, encapsulation efficiency, and zeta potential; performed cell culture to eventually test therapeutic on RAW cells; used an ELISA assay and MTT assay for cytokine expression analysis and cell viability data.
- Project II:
  - Involved with the development of a novel model of Fabry Disease using iPSCs to differentiate disease-specific fibroblasts into cardiomyocytes
  - Worked on a novel application of L-Arginine, nano-L-Arginine, and its role in decreasing oxidative stress and mitochondrial ROS in Fabry patients.
  - Studied the specific differences in cell morphology between healthy patients and Fabry patients to understand the molecular mechanisms behind cardiovascular morbidity.

WORK EXPERIENCE

Cardiovascular Shadowing at the Private Practice of Dr. Sachin Jain **June 2021- August 2022**
Medical Shadow

- Shadowed an interventional cardiologist, along with his team of doctors, nurses, and medical assistants
- Observed sclerotherapy; helped perform services such as EKG and API; observing patient-doctor interactions
- Responsible for taking vitals and supporting nursing staff with patient care in the recovery room; assisted with translation services

STARTALK **June 2021- August 2021**
Student Assistant

- Assisted a Hindi-language teacher during a one-month summer camp; created lesson plans focused on culture, language and literature; responsible for grading and the delivery of instruction to Indian-American children in grades 1-4

Urja Jain Tutoring **June 2022- Present**
Independent Tutor

- Supported students in STEM subjects to improve their understanding of complex topics
- Organized personalized sessions in Pre-Algebra, Geometry, and laboratory science; improved academic performance by over 20%.

Baking Small Business **December 2023- Present**
CEO and Founder

- Founded a sustainable small business to create vegan alternatives and to educate consumers on the importance of sustainable eating and living; $1000+ in sales
- Crafted personalized, eco-friendly vegan baked goods, such as cakes and pastries
- Supplied 10+ bake sales and donations to shelters

EDUCATION

Ardsley High School
High School Diploma awarded June 2025, GPA: 4.1/4.0 (UW)

Columbia University
High School Science Honors Program
Accepted into a highly selective program for high school students taught by Columbia faculty and graduate students in Bioengineering and Biotechnology, focusing on tissue engineering and novel developments in the field.

University of California, Berkeley
College diploma to be awarded May 2029

HONORS AND AWARDS

- National Honor Society and French Honor Society
- Le Grand Concours National Silver and Bronze medals (x2)
- DECA Region 3 Conference (2nd place, 3rd place)
- DECA NY State Conference (Top 10 Written and Overall x3)
- Somers Science Fair 2nd place
- Science Olympiad Regionals (2nd and 3rd place)
- Westchester Science & Engineering Fair (WESEF) 1st & 3rd
- International GENIUS Finalist and Honorable Mention
- JSHS Finalist and State Presenter
- Future of Medicine Award
- Ardsley Congress of Teachers (ACT) Scholarship
- Excellence in French Language and Seal of Biliteracy